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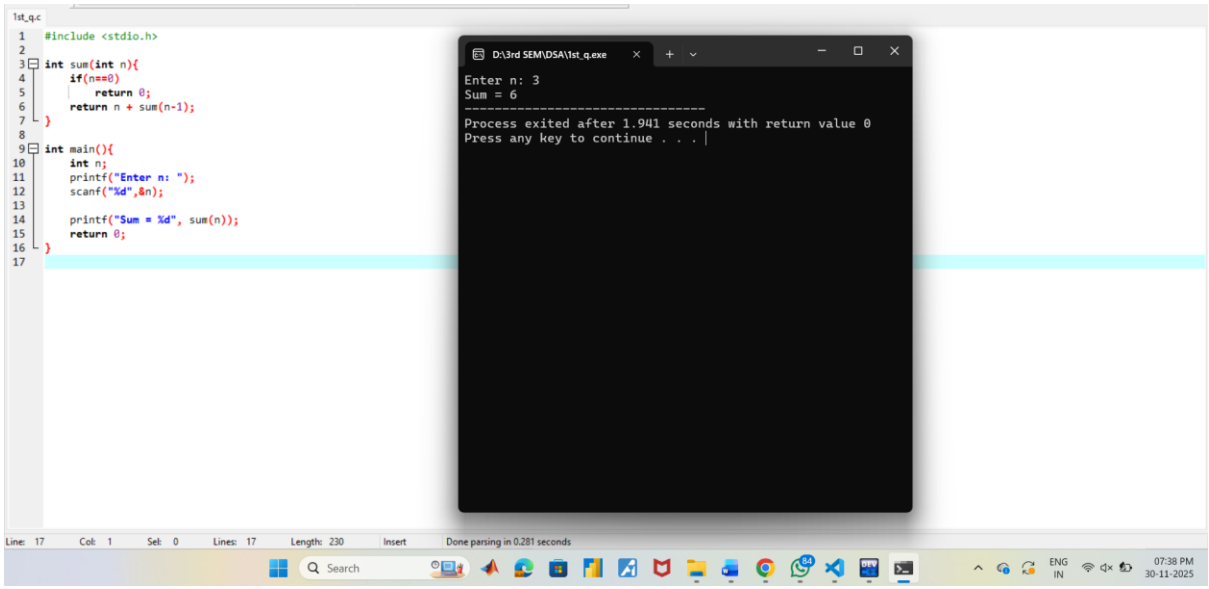
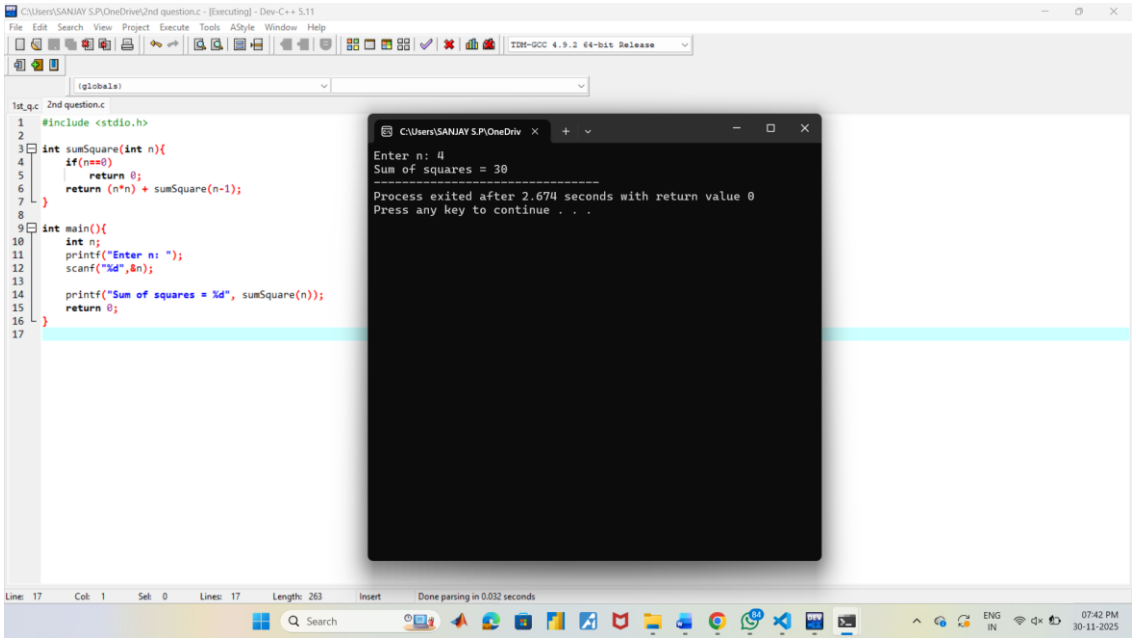
2nd Year CSE-C

QUESTION 1



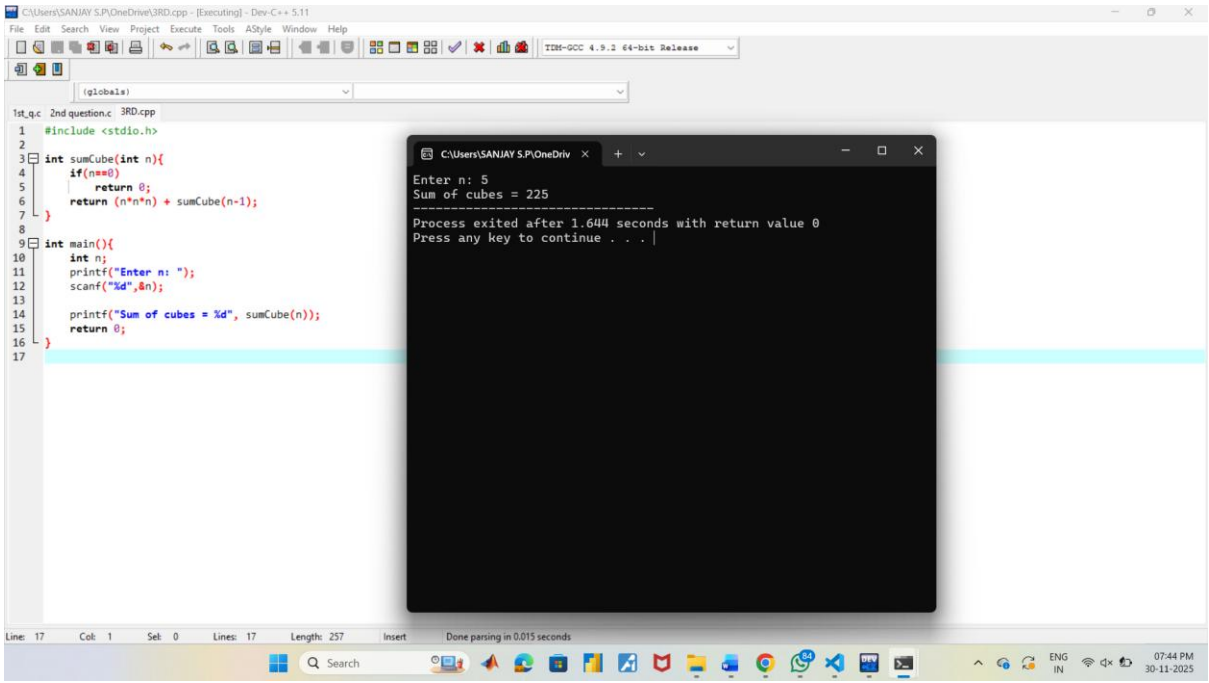
This code is telling us the sum of n numbers in recursion and one by one process of implementation till the condition fail.

QUESTION 2



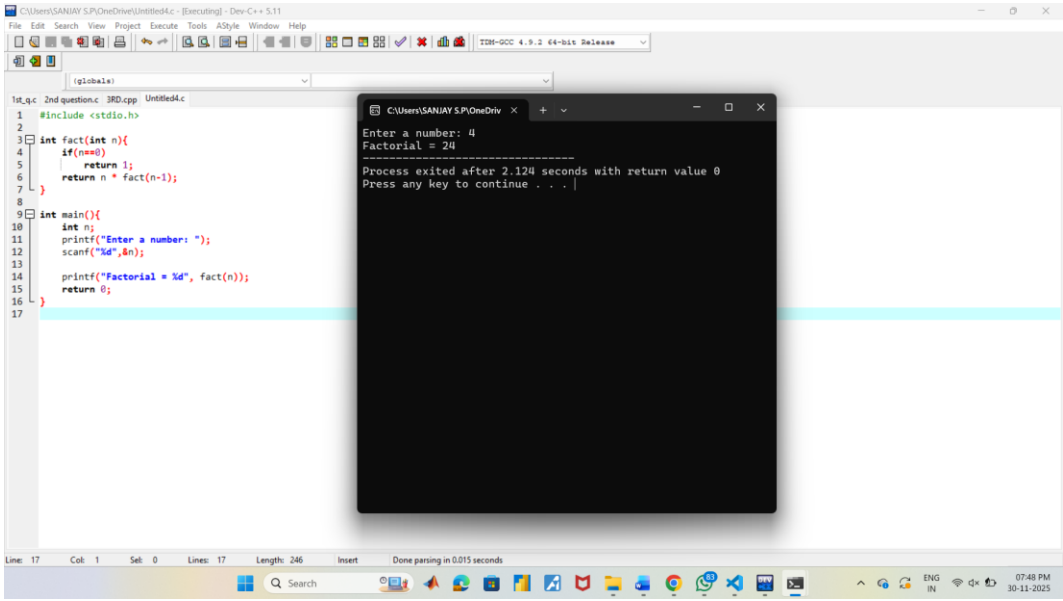
Like last question this also does the same operation in terms of squares.

Question 3



Like last question this also does the same operation in terms of cubes.

Question-4



Like in this code we are doing the factorial operation and it works with recursion and it exits the code at termination stage

Question 5

```c
#include <stdio.h>

int a[3][3];

void transpose(int i, int j){
    if(i==3) return;
    printf("%d ", a[j][i]);
    if(j==2){
        printf("\n");
        transpose(i+1,0);
    }
    else
        transpose(i,j+1);
}

int main(){
    int i,j;
    printf("Enter 3x3 matrix elements:\n");
    for(i=0;i<3;i++)
        for(j=0;j<3;j++)
            scanf("%d",&a[i][j]);

    printf("Transpose of matrix:\n");
    transpose(0,0);

    return 0;
}
```

```
Enter 3x3 matrix elements:
3
4
5
6
7
8
9
2
3
Transpose of matrix:
3 6 9
4 7 2
5 8 3

--------------------------------
Process exited after 6.987 seconds with return value 0
Press any key to continue . . .
```

It's the transposes of the 3X3 matrix and using the input it transposes the matrix

Question 6:

```c
#include <stdio.h>

int fib(int n){
    if(n<=1)
        return n;
    return fib(n-1) + fib(n-2);
}

int main(){
    int n;
    printf("Enter number of terms: ");
    scanf("%d",&n);

    printf("Fibonacci series:\n");
    for(int i=0;i<n;i++){
        printf("%d ", fib(i));
    }
    return 0;
}
```

```
Enter number of terms: 4
Fibonacci series:
0 1 1 2
--------------------------------
Process exited after 2.594 seconds with return value 0
Press any key to continue . . .
```

this code uses the Fibonacci method like using recursive function and using n-1 and n-2 it take the previous digits for calculation .

THANK YOU